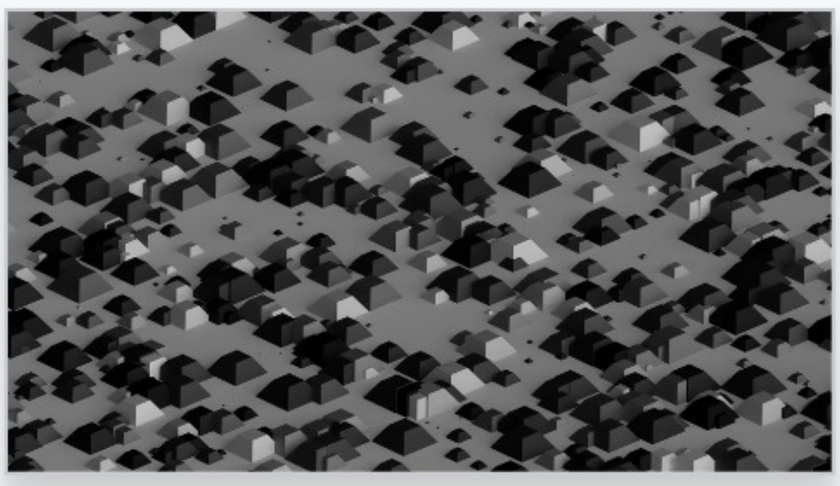
V1.0 SCHEMATIC SYSTEM

Explore Biology Through the Lens of a Bustling City.

Cell City is a high-fidelity educational simulation where biological processes are mapped to urban infrastructure. Decode genomes, manage metabolic power grids, and stabilize protein factories.

Enter the City →

Open Bio Lab



ORGANELLES
12+ Districts

CHALLENGES
03 Scenarios

STATUS
• Operational

Interactive Map
Browse the organelle districts.

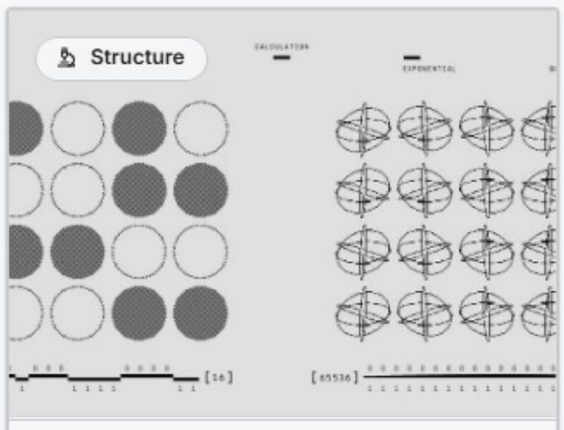
Bio Lab
Run DNA sequence diagnostics.

Challenges
Solve active cellular crises.

KNOWLEDGE MODULES

Explore the Cellular Ecosystem

Master the fundamental systems that drive cellular life. Each module provides interactive data visualizations and comparative structural analysis.



Cellular Architecture

Analyze the structural blueprints of eukaryotic and prokaryotic systems. Understand how the 'city walls' and

Explore Module →



Metabolic Power Grids

Dive into the mitochondria power plants. Learn how ATP is generated through complex biochemical assembly lines to

Explore Module →



Inter-City Communication

Decipher the signal transduction pathways that allow cells to respond to their environment. Explore receptors,

Explore Module →

Ready to stabilize the system?

Join thousands of biology students and researchers in our virtual city environment. Solve real-world crisis scenarios and develop mastery of cellular biology.

Get Started Free

Contact Support

Cell City Project

An immersive, educational platform where students explore cellular biology by conceptualizing the cell as a sophisticated, interconnected urban infrastructure.

Resources

- Educator Portal
- Curriculum Guide
- Privacy Policy

Support

- support@cellcity.edu
- FAQ & Troubleshooting
- Community Forum

Legal

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System Visualization

Cell City: Interactive Map

Explore the cellular landscape through the lens of urban infrastructure. Click on city districts to investigate organelle functions and discover how biological systems mirror human engineering.

City Directory

Select a district to view technical specs

NUCLEUS
CITY HALL & DATA CENTER

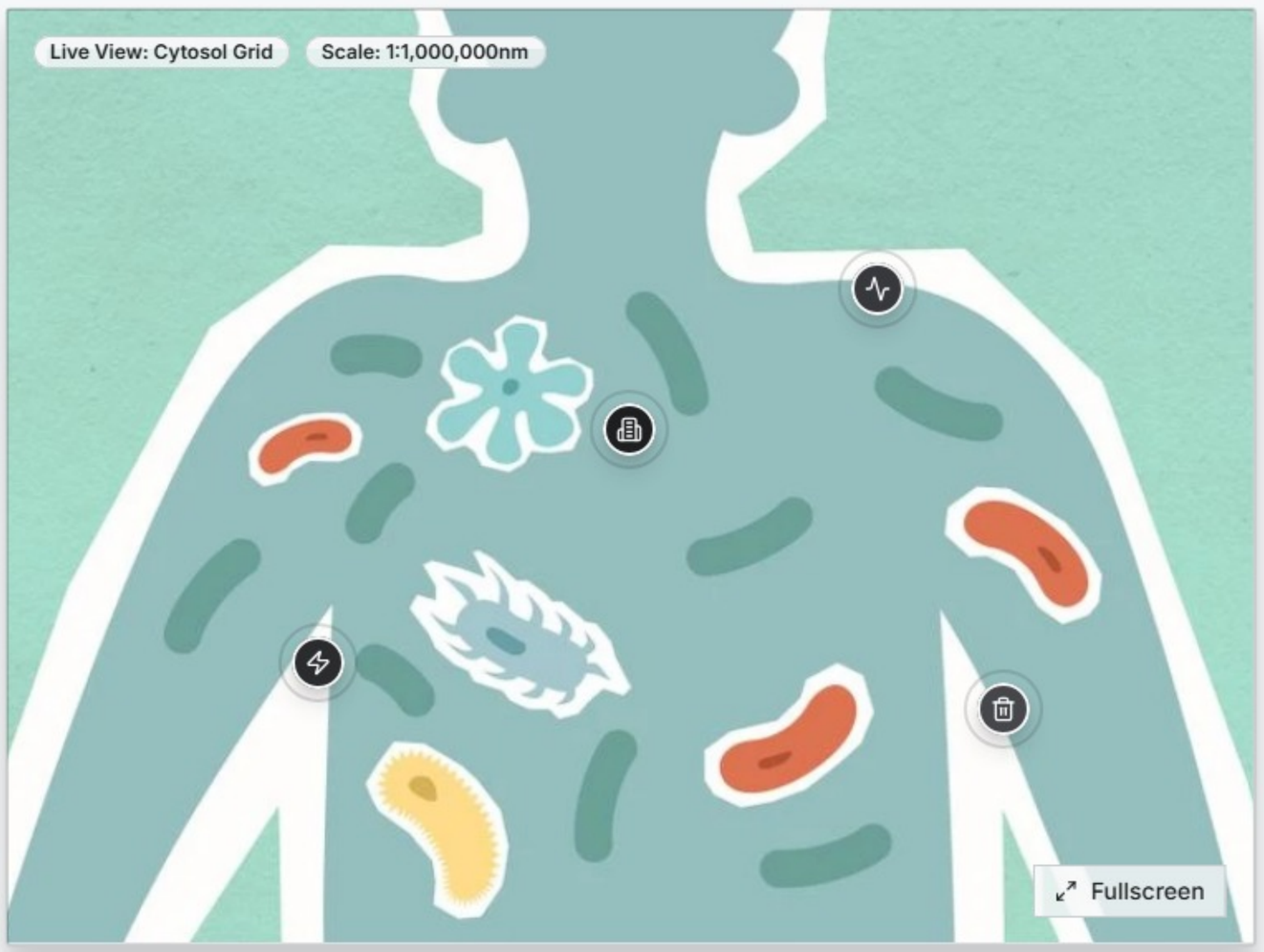
MITOCHONDRIA
POWER PLANT

RIBOSOMES
MANUFACTURING UNITS

LYSOSOMES
WASTE MANAGEMENT & RECYCLING

Navigation Tip

Hover over hotspots on the map to highlight organelles. Click to expand full biological functional descriptions.



NUCLEUS | MITOCHONDRIA | RIBOSOMES | LYSOSOMES

Why a City?

Cells are not static bags of chemicals; they are highly organized metabolic machines. By comparing organelles to urban infrastructure, we can better understand the logistical challenges of life: energy production, waste disposal, communication, and manufacturing.

- Centralized vs. Distributed systems
- Energy conversion and efficiency
- Information flow (DNA to Protein)
- Internal transportation and structural support



Read the Theory | Bio Lab Intro

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LABORATORY CORE

Tool Selection

DNA Sequencing
Analyze and manipulate raw DNA base pairs.

mRNA Translation
Convert DNA to mRNA and synthesize protein chains.

Codon Interpreter
Map nucleotide triplets to their respective amino acids.

Mutation Compare
Identify differences between wild-type and mutant strands.

>_ LAB STATUS: ONLINE

"The cell is a programmable city. DNA is its fundamental source code."

BIOINFORMATICS SECTOR

Digital Lab Terminal

Process biological data through the city's central processing hub. Manipulate sequences, translate genetic instructions, and diagnose cellular anomalies.

Export Data | Analyze System

MODULE 01

Sequence Analysis

>_ Input Buffer

Enter DNA nucleotide sequence for validation and basic metrics.

```
ATGCGTACGTAGCTAGCTAGCATGCTAGCTAGCTAGCTAGCTAGCTAGCTAGCTAGCT
```

Clear | Randomize | Run Diagnostics

NUCLEOTIDE DISTRIBUTION

Adenine (A)	18 bps
Thymine (T)	16 bps
Cytosine (C)	14 bps
Guanine (G)	16 bps

GC CONTENT **47%**

Sequence Preview

```
1           11          21          31          41          51          61
ATGCGTACGT  AGCTAGCTAG  CATGCTAGCT  AGCTAGCTAG  CTAGCTAGCT  AGCTAGCTAG  CT
```

PROTOCOL LOGS

SESSION AUTHORIZATION

User verified via Cell City Educational ID. Write-access granted for Lab Module A-12.

SECURITY STANDARDS

ENVIRONMENT TELEMETRY

Temp Stability	37.0°C
pH Calibration	7.40
Reagent Level	82%

Need Lab Assistance?

Review the Bioinformatics manual for advanced terminal commands.

View Tutorial

Made with Visily

SYSTEM ALERT: ACTIVE INCIDENTS

City Challenges

The Cell City is experiencing critical system failures. As the Lead Systems Biologist, you must analyze the data, identify the metabolic bottlenecks, and restore cellular homeostasis across all districts.

Active Scenarios | Resolved | Archives

12/15 Scenarios Cleared

84% Average Efficiency

Challenge Guide



MITOCHONDRIA DISTRICT

25 min

Energy Crisis

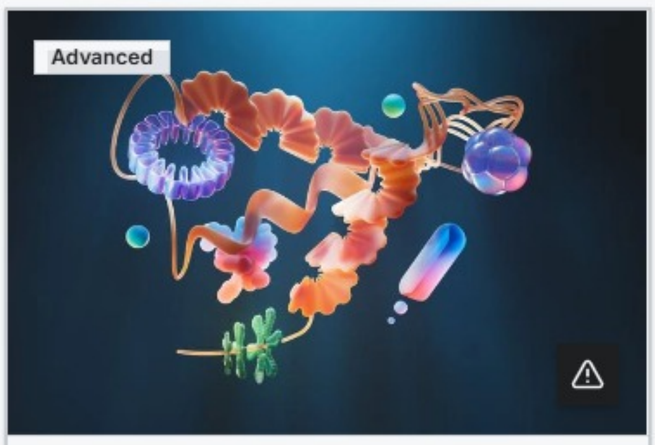
The mitochondrial power plants are failing. ATP production has dropped by 60%, threatening basic city functions like transport and synthesis.

PRIMARY REWARD 500 XP

CURRENT STATUS UNSOLVED

STABILITY LEVEL CRITICAL

DEPLOY RESPONSE →



RIBOSOMES & ER DISTRICT

40 min

Protein Factory

Demand for structural proteins has spiked. Optimize the Ribosome assembly lines and Endoplasmic Reticulum logistics to prevent a structural collapse

PRIMARY REWARD 850 XP

CURRENT STATUS UNSOLVED

STABILITY LEVEL CRITICAL

DEPLOY RESPONSE →



LYSOSOMES DISTRICT

15 min

Waste Crisis

Lysosome recycling centers are overwhelmed by metabolic debris. Clear the backlog before toxicity levels trigger an apoptosis (self-destruct) sequence

PRIMARY REWARD 300 XP

CURRENT STATUS UNSOLVED

STABILITY LEVEL CRITICAL

DEPLOY RESPONSE →

Need Laboratory Support?

Before attempting higher-level challenges, ensure your understanding of cellular machinery is complete. Use the Bioinformatics Lab to analyze specific sequences or revisit the Interactive City map to locate critical infrastructure.

Open Bio Lab | View City Map

SCENARIO LOG

- Energy output stabilized in Nucleus — 2h ago
- New protein synthesis task added — 5h ago
- Waste backlog alert in Lysosome — 1d ago

Upcoming Scenarios

Unlock these by completing the foundational challenges above.

GENETIC DRIFT
Analyze mutations in the core database. →

MEMBRANE BREACH
Repair the perimeter security system. →

VIRAL INCURSION
Defend the city against external threats. →

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